COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	07/16/2026	**DATE SUBMITTED:**	07/16/2026
NAME:	Charles Vence Espinas	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 1.1 **Creating Virtual Machine**

1. PROCEDURE OUTPUT

Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

1. Change the hostname using the command sudo nano /etc/hostname
1.1 Use server1 for Server 1
1.2 Use server2 for Server 2
1.3 Use workstation for the Local Machine

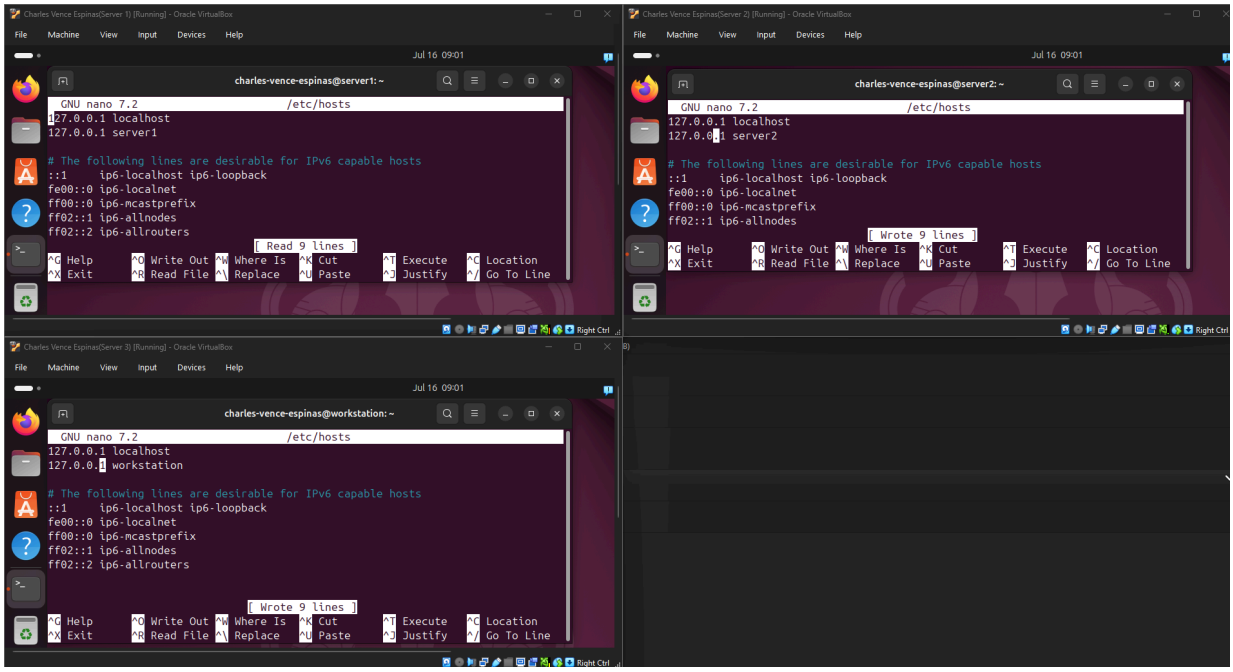
```
charles-vence-espinas@server1:~$ sudo nano /etc/hostname
[sudo] password for charles-vence-espinas:
charles-vence-espinas@server1:~$ cat /etc/hostname
server1
```

```
charles-vence-espinas@charles-vence-espinas-VirtualBox:~$ sudo nano /etc/hostname
[sudo] password for charles-vence-espinas:
charles-vence-espinas@charles-vence-espinas-VirtualBox:~$ cat /etc/hostname
server2
```

```
charles-vence-espinas@charles-vence-espinas-VirtualBox:~$ sudo nano /etc/hostname
[sudo] password for charles-vence-espinas:
charles-vence-espinas@charles-vence-espinas-VirtualBox:~$ cat /etc/hostname
workstation
```

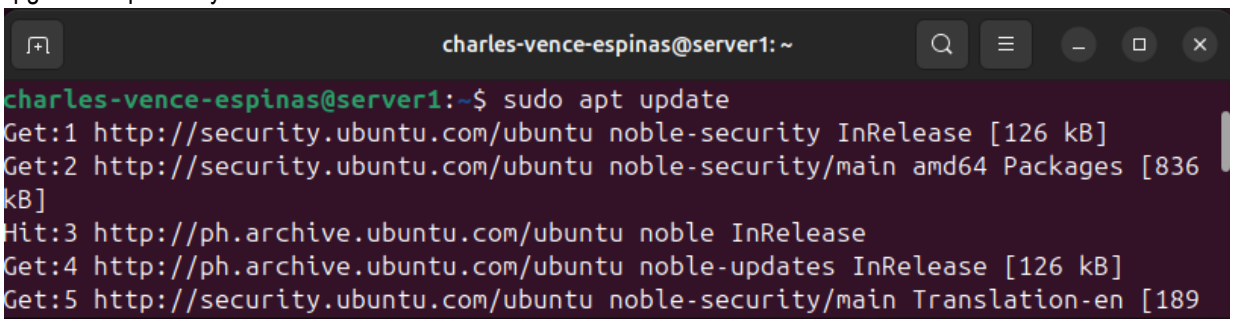
-In this part, I change the hostname of three machine by using the sudo nano /etc/hostname

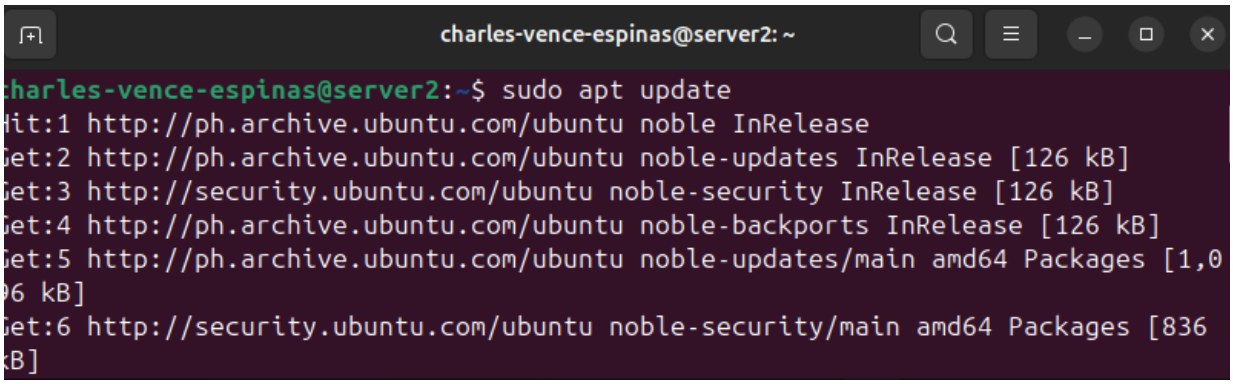
2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.
2.1 Type 127.0.0.1 server 1 for Server 1
2.2 Type 127.0.0.1 server 2 for Server 2
2.3 Type 127.0.0.1 workstation for the Local Machine

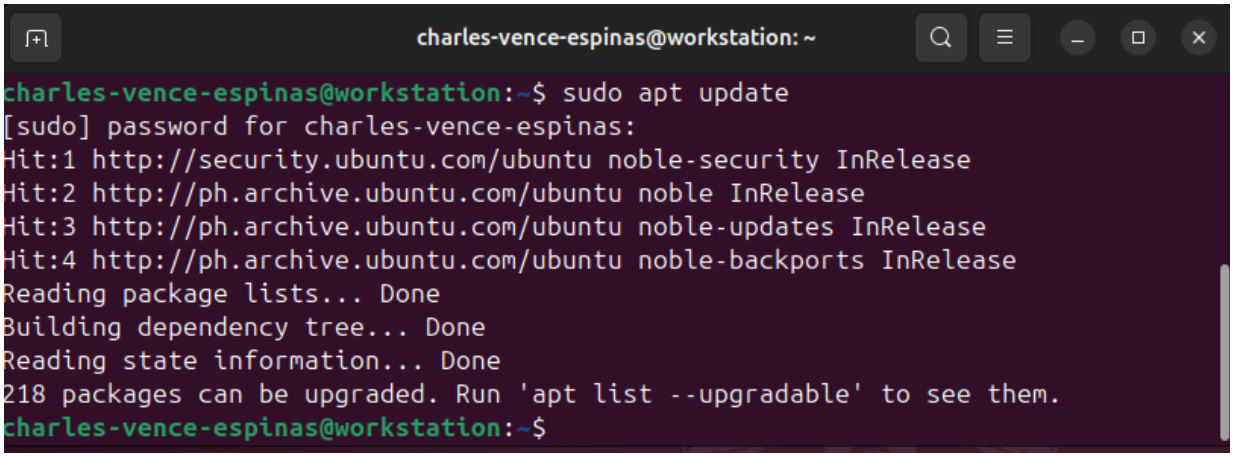


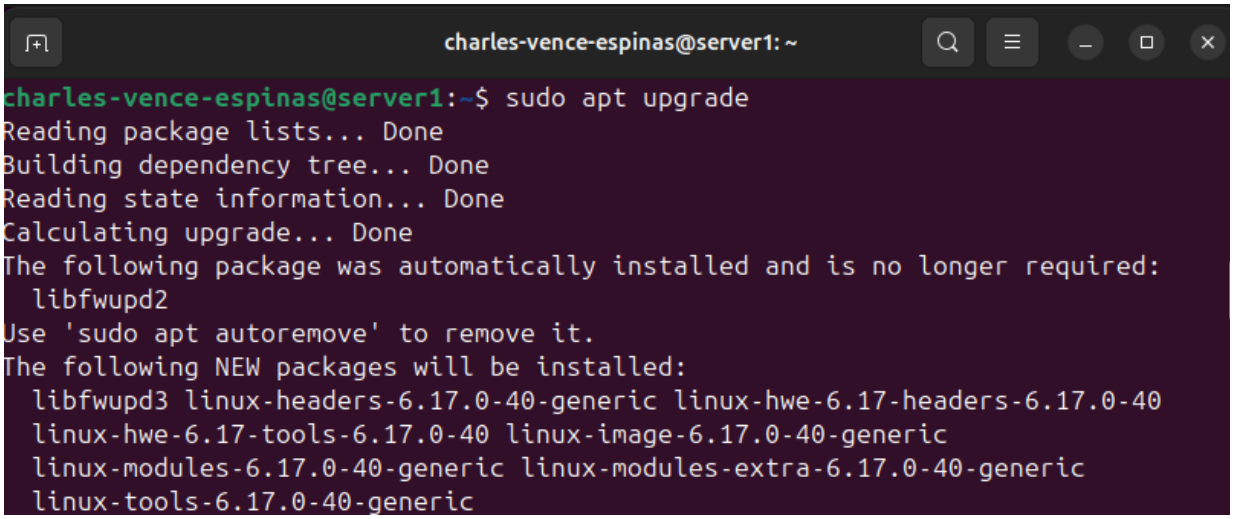
-In this part,I modify the file/etc/hosts by adding the ip and the hostname of three machine

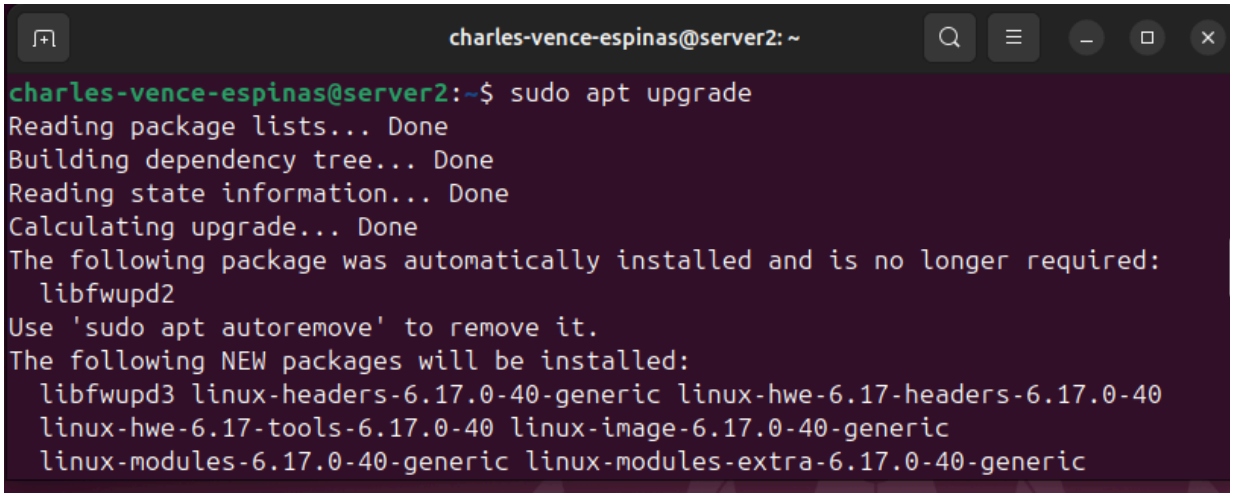
Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:
1. Upgrade the packages by issuing the command sudo apt update and sudo apt upgrade respectively.

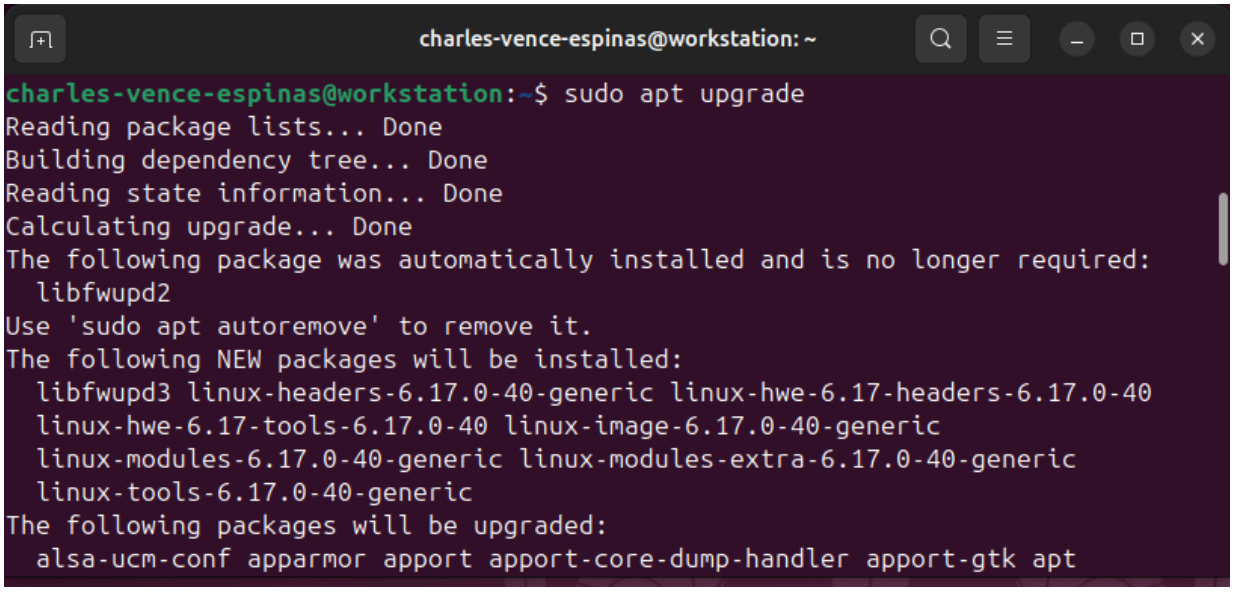




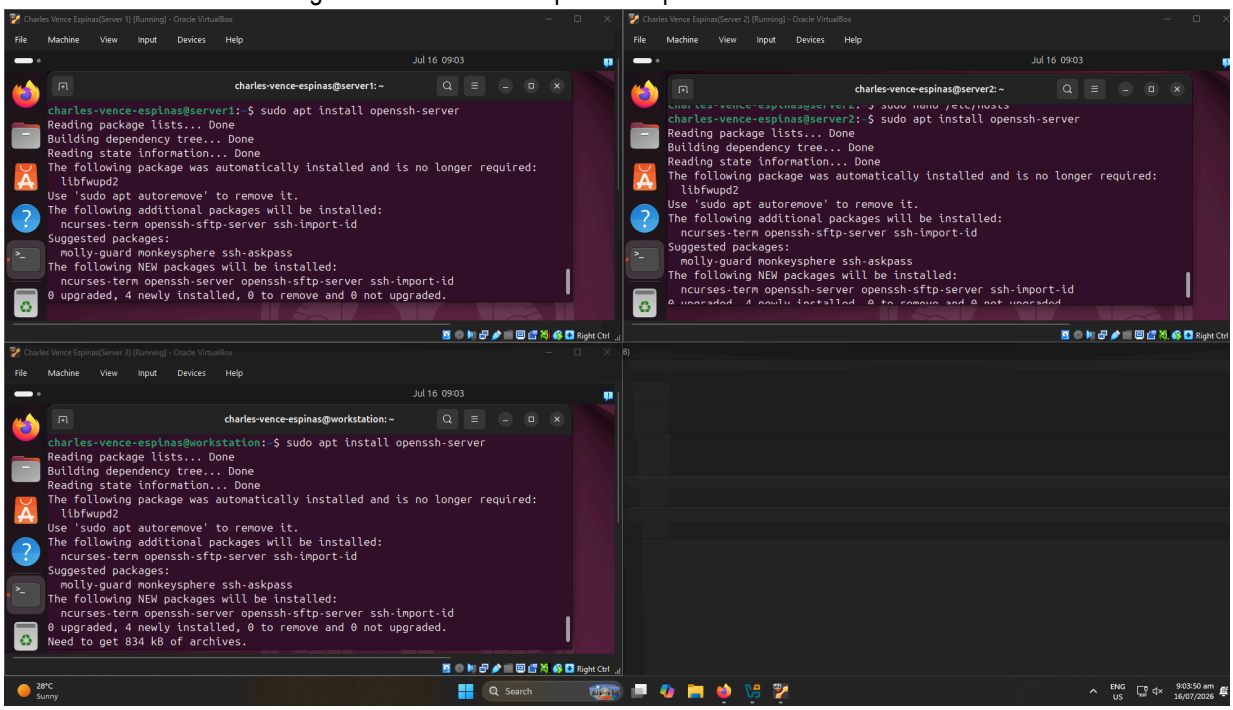

charles-vence-espinas@workstation: ~
charles-vence-espinas@workstation:~$ sudo apt update
[sudo] password for charles-vence-espinas:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
218 packages can be upgraded. Run 'apt list --upgradable' to see them.
charles-vence-espinas@workstation:~$


charles-vence-espinas@server1: ~
charles-vence-espinas@server1:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
libfwupd2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
libfwupd3 linux-headers-6.17.0-40-generic linux-hwe-6.17-headers-6.17.0-40
linux-hwe-6.17-tools-6.17.0-40 linux-image-6.17.0-40-generic
linux-modules-6.17.0-40-generic linux-modules-extra-6.17.0-40-generic
linux-tools-6.17.0-40-generic


charles-vence-espinas@server2: ~
charles-vence-espinas@server2:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
libfwupd2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
libfwupd3 linux-headers-6.17.0-40-generic linux-hwe-6.17-headers-6.17.0-40
linux-hwe-6.17-tools-6.17.0-40 linux-image-6.17.0-40-generic
linux-modules-6.17.0-40-generic linux-modules-extra-6.17.0-40-generic

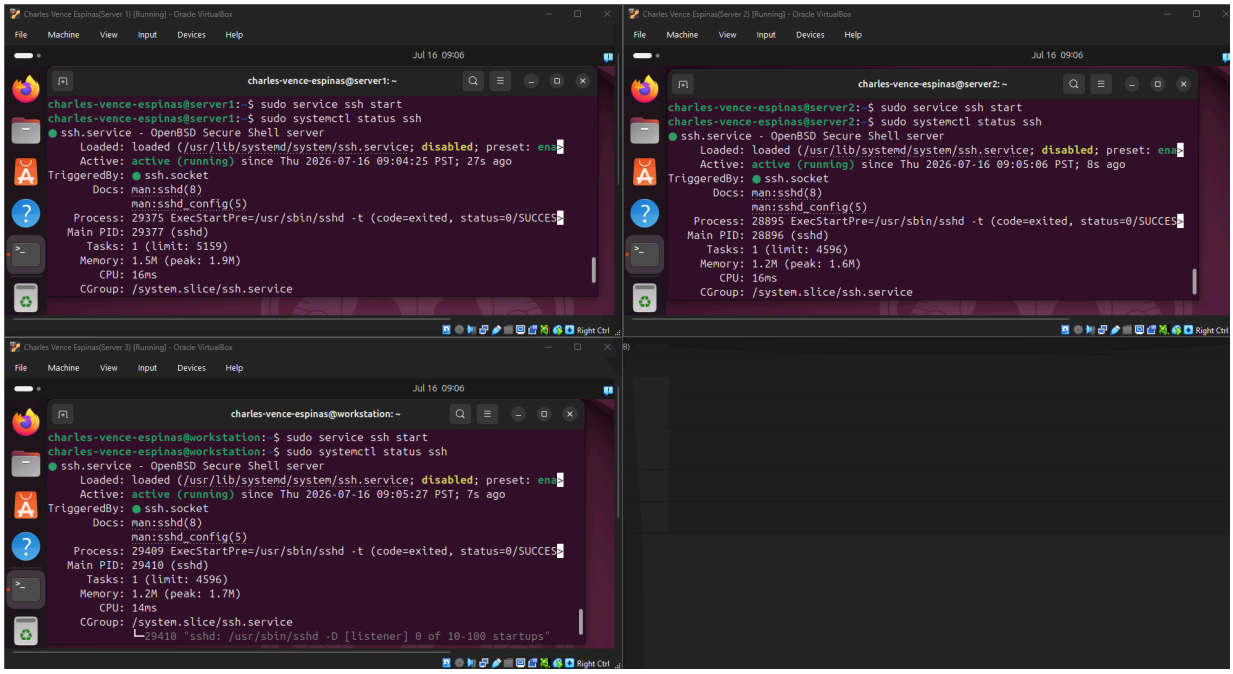


2. Install the SSH server using the command sudo apt install openssh-server.



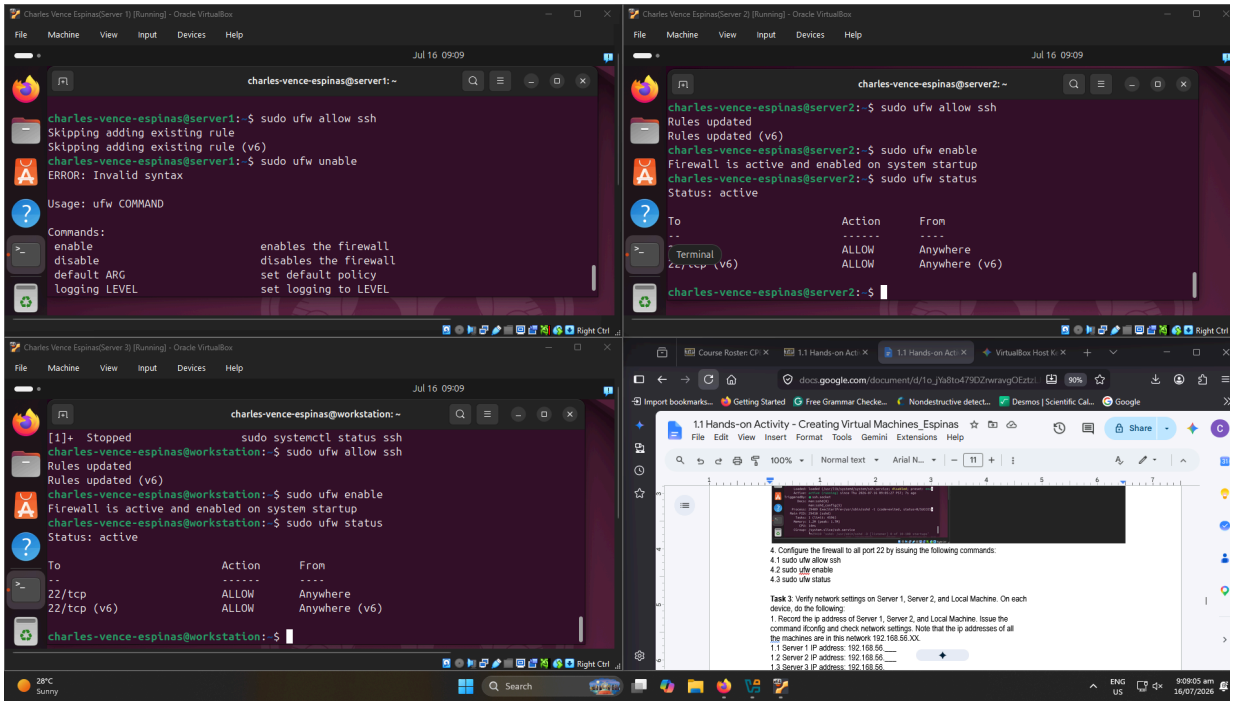
-I install and upgrade the system in three machine

3. Verify if the SSH service has started by issuing the following commands:
3.1 sudo service ssh start
3.2 sudo systemctl status ssh



-I install the ssh service in three machines

4. Configure the firewall to all port 22 by issuing the following commands:
4.1 sudo ufw allow ssh
4.2 sudo ufw enable
4.3 sudo ufw status



-In this part, I configure a firewall for securing a new linux server.

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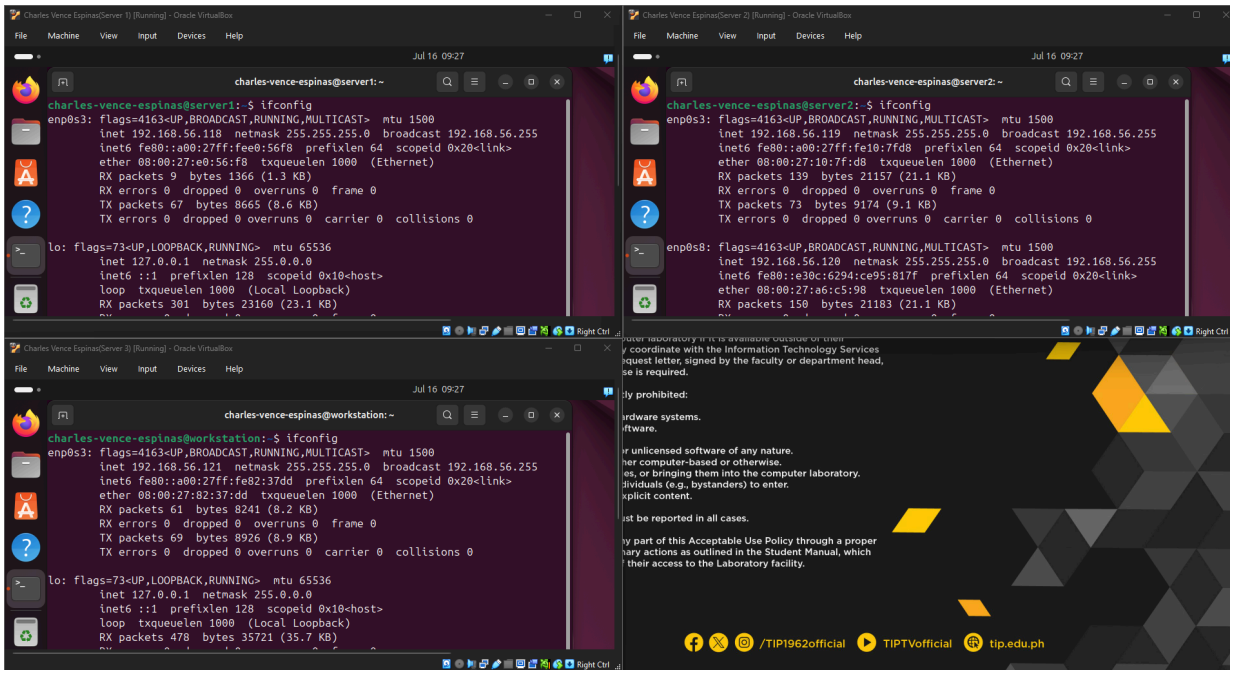
Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.

1.1 Server 1 IP address: 192.168.56.118

1.2 Server 2 IP address: 192.168.56.119

1.3 Server 3 IP address: 192.168.56.121



-in this part, I use ifconfig to show the ip address of each machine.

2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: **/ Successful** ☐ Not Successful

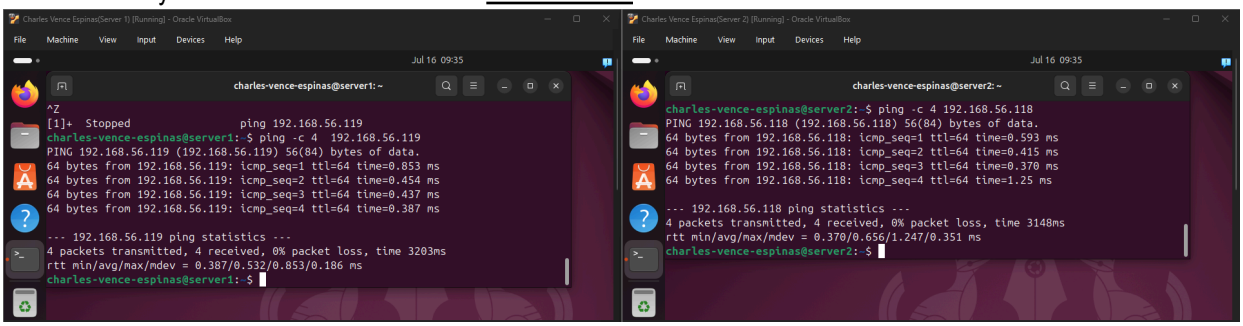
2.2 Connectivity test for Local Machine 1 to Server 2: **/ Successful** ☐ Not Successful

```
charles-vence-espinas@workstation:~$ ping -c 4  192.168.56.118
PING 192.168.56.118 (192.168.56.118) 56(84) bytes of data.
64 bytes from 192.168.56.118: icmp_seq=1 ttl=64 time=0.673 ms
64 bytes from 192.168.56.118: icmp_seq=2 ttl=64 time=0.354 ms
64 bytes from 192.168.56.118: icmp_seq=3 ttl=64 time=0.371 ms
64 bytes from 192.168.56.118: icmp_seq=4 ttl=64 time=0.392 ms

--- 192.168.56.118 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3281ms
rtt min/avg/max/mdev = 0.354/0.447/0.673/0.130 ms
```

```
charles-vence-espinas@workstation:~$ ping -c 4 192.168.56.119
PING 192.168.56.119 (192.168.56.119) 56(84) bytes of data.
64 bytes from 192.168.56.119: icmp_seq=1 ttl=64 time=0.816 ms
64 bytes from 192.168.56.119: icmp_seq=2 ttl=64 time=0.650 ms
64 bytes from 192.168.56.119: icmp_seq=3 ttl=64 time=0.489 ms
64 bytes from 192.168.56.119: icmp_seq=4 ttl=64 time=0.420 ms
```

2.3 Connectivity test for Server 1 to Server 2: **/ Successful** ☐ Not Successful



-In this part, I test the connection of three devices.

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120

1.2 Enter the password for server 1 when prompted

1.3 Verify that you are in server 1. The user should be in this format user@server1. For example, jvtaylar@server1

```
charles-vence-espinas@workstation:~$ ssh charles-vence-espinas@192.168.56.118
The authenticity of host '192.168.56.118 (192.168.56.118)' can't be established
ED25519 key fingerprint is SHA256:mLcdPmyO+xxwwdnliUSYaKmLil1azLLIa8JPD02XsBw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.118' (ED25519) to the list of known host
.
charles-vence-espinas@192.168.56.118's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

-In this part, I enter the server 1 from my local machine pc using ssh username@ipaddressOfServer1

2. Logout of Server 1 by issuing the command control + D.

```
charles-vence-espinas@server1:~$
logout
Connection to 192.168.56.118 closed.
```

3. Do the same for Server 2.

```
charles-vence-espinas@workstation:~$ ssh charles-vence-espinas@192.168.56.119
The authenticity of host '192.168.56.119 (192.168.56.119)' can't be established.
ED25519 key fingerprint is SHA256:mLcdPmyO+xxwwdnliUSYaKmLil1azLLIa8JPD02XsBw.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.119' (ED25519) to the list of known hosts
.
charles-vence-espinas@192.168.56.119's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

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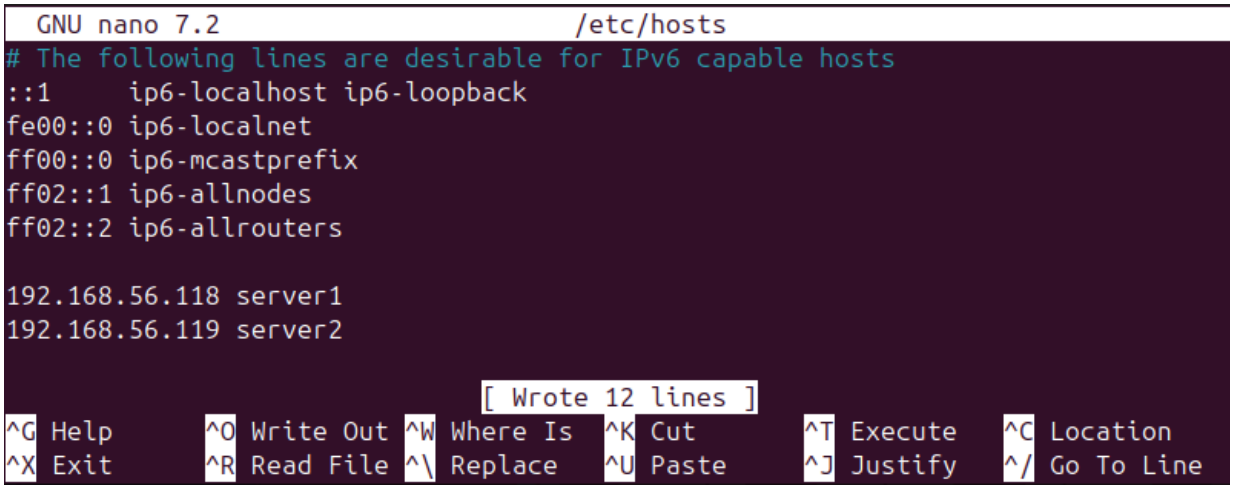

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applicable law.

charles-vence-espinas@server2:~$
```

I also enter the server2 using the ssh command ssh username@ipaddressOfServer2.

4. Edit the hosts of the Local Machine by issuing the command sudo nano /etc/hosts. Below all texts type the following:
4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)
4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)
4.3 Save the file and exit.



-I enter the ip address and hostname of the 2 server from the local machine /etc/hosts so that I can use either the ipaddress or the hostname in the command ssh username@ipaddress

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2

```
charles-vence-espinas@workstation:~$ ssh charles-vence-espinas@server1
The authenticity of host 'server1 (192.168.56.118)' can't be established.
ED25519 key fingerprint is SHA256:mLcdPmyO+xxwwdnliUSYaKmLil1azLLIa8JPD02XsBw.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known hosts.
charles-vence-espinas@server1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

Last login: Thu Jul 16 09:38:08 2026 from 192.168.56.121
charles-vence-espinas@server1:~$ hostname
server1
```

```
charles-vence-espinas@workstation:~$ ssh charles-vence-espinas@server2
The authenticity of host 'server2 (192.168.56.119)' can't be established.
ED25519 key fingerprint is SHA256:mLcdPmyO+xxwwdnliUSYaKmLil1azLLIa8JPD02XsBw.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
    ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yeds
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'server2' (ED25519) to the list of known hosts.
charles-vence-espinas@server2's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

Last login: Thu Jul 16 09:41:08 2026 from 192.168.56.121
charles-vence-espinas@server2:~$ hostname
server2
```

-In this part, I enter the 2 servers pc using the ssh command and their username using ssh username@hostname.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?

We were able to use the hostname instead of IP address in SSH commands by using the sudo nano /etc/hosts and then we modified that file and added the ip address and the hostname of the server. In this way, we were able to specify to the system that we can use either an ip address or the hostname of the computer using SSH commands.

2. How secured is SSH?

Secure shell is highly secured because it can encrypt all the information and data about the communication between devices and servers SSH also supports authentication which means that it will require a password of the device that will be entered in of other devices which the authorized users can only be entered and it can help to be more secured to the attacker or hackers.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic

work. The following indicates the nature and extent of AI assistance:

Tools Used: **Chatgpt, Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **For searching facts about something and grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I use this tool so that I can find facts about secure shell**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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